



VOLT EDGE

Strait of Hormuz Crisis

Multi-Horizon Policy Roadmap: Immediate
Stability, Diversification, Thorium-Led
Independence

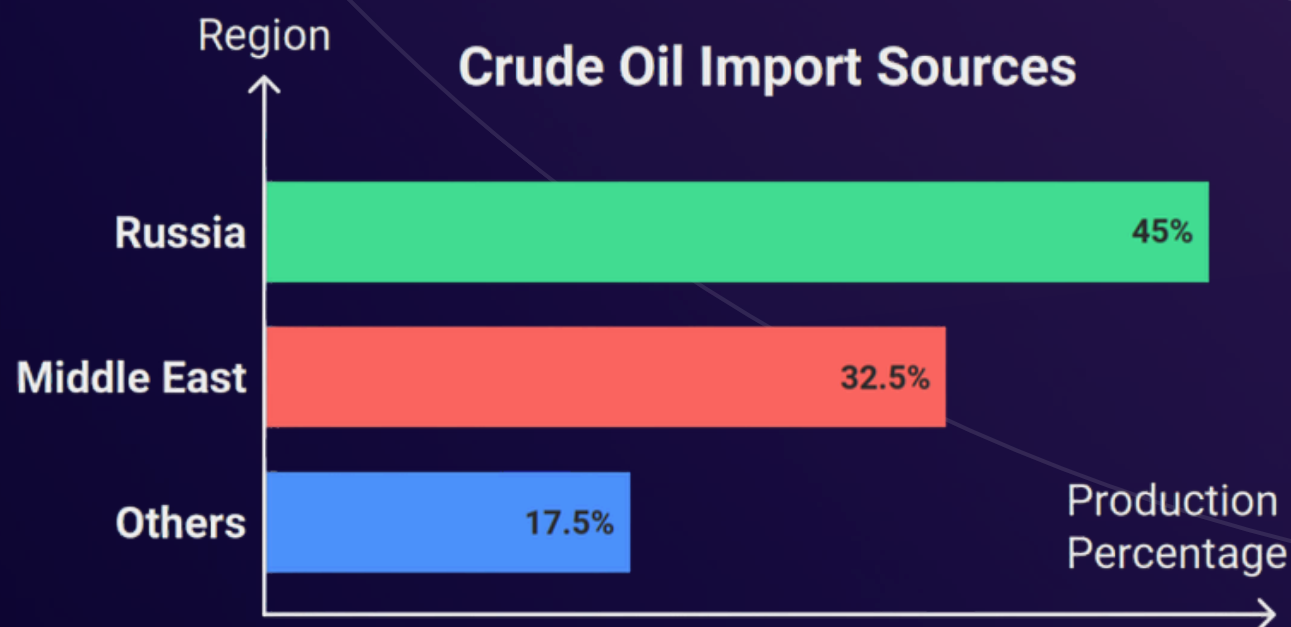
Contain the Shock – Diplomacy, Reserves, Russia Pivot & Domestic Priority



✔ Immediate Measures:

- **Diplomatic & Safe Passage:** Build on MEA-Iran safe passage for Indian vessels + QUAD/US naval escorts; secure waivers for non-Hormuz routes.
- **SPR Utilization & Acceleration:** Phased drawdown from current 74-day total buffer (5.33 MMT caverns + OMC/refiner stocks); fast-track Phase-II (Chandikhol/Padur +6.5 MMT) for added 12 days.
- **Discounted Russian Surge + Alternatives:** Ramp Russia to 40–50%+ share (current surge: 1.5–2 mbpd via US waivers; 30M barrels snapped up weekly); emergency spot from US, West Africa, Latin America.
- **Domestic Priority & Demand Side:** Invoke export restrictions/priority allocation for petroleum products (refineries redirect to domestic under Essential Commodities Act); targeted subsidies for LPG/fertiliser; voluntary conservation + partial price pass-through.

Categorie s	Unmitigated	Mitigated
GDP	-0.7 to -1.0 pp	-0.3 to -0.6 pp
Inflation	+4.0 to +5.5 pp	+3.5 to +4.0 pp
CAD	+1.0 to +1.5% GDP	+0.5–0.8% GDP
Rupee	8–12% depreciation	4–7% depreciation



✔ Economic Impact:

- Oil spike \$110–130/bbl → **Unmitigated:** GDP ↓ 0.7–1 pp, inflation ↑ 4–5.5%, CAD +1–1.5% GDP, rupee 8–12% depreciation.
- **Mitigated** (Russia pivot + SPR + exports redirect): GDP loss capped 0.3–0.6 pp, inflation ~3.5–4%, import bill contained via discounts.
- **Fiscal cost:** ₹18,000–25,000 Cr (subsidies + SPR refill) → reallocate from cess/export duties.

Reduce Dependence – Accelerate RE, Nuclear, & Non-ME Supplies

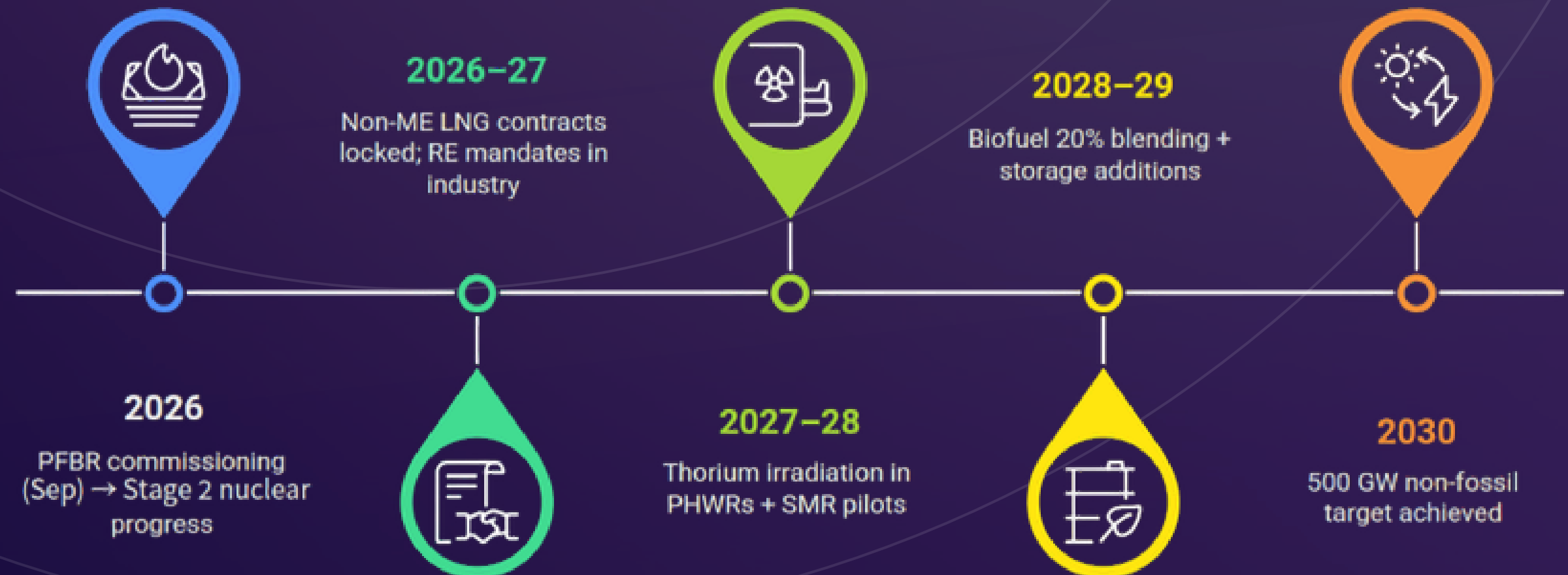


✓ Key Insights

- **LNG/Oil Diversification:** Secure 5–10 year contracts with US/Canada/Australia (non-ME LNG share ↑ to 65%+); expand terminals + overland options.
- **Renewable Fast-Track:** Push to 500 GW non-fossil by 2030 (already ~52% achieved early); mandate 50% RE in new industry/CGD; add storage (41 GW target).
- **Nuclear Acceleration:** Commission PFBR (Sep 2026) for Stage 2; fast-track thorium irradiation in PHWRs + SMRs/private via SHANTI Act.
- **Domestic & Biofuels:** ONGC exploration incentives; 20% biofuel blending mandate.

Climate Alignment:

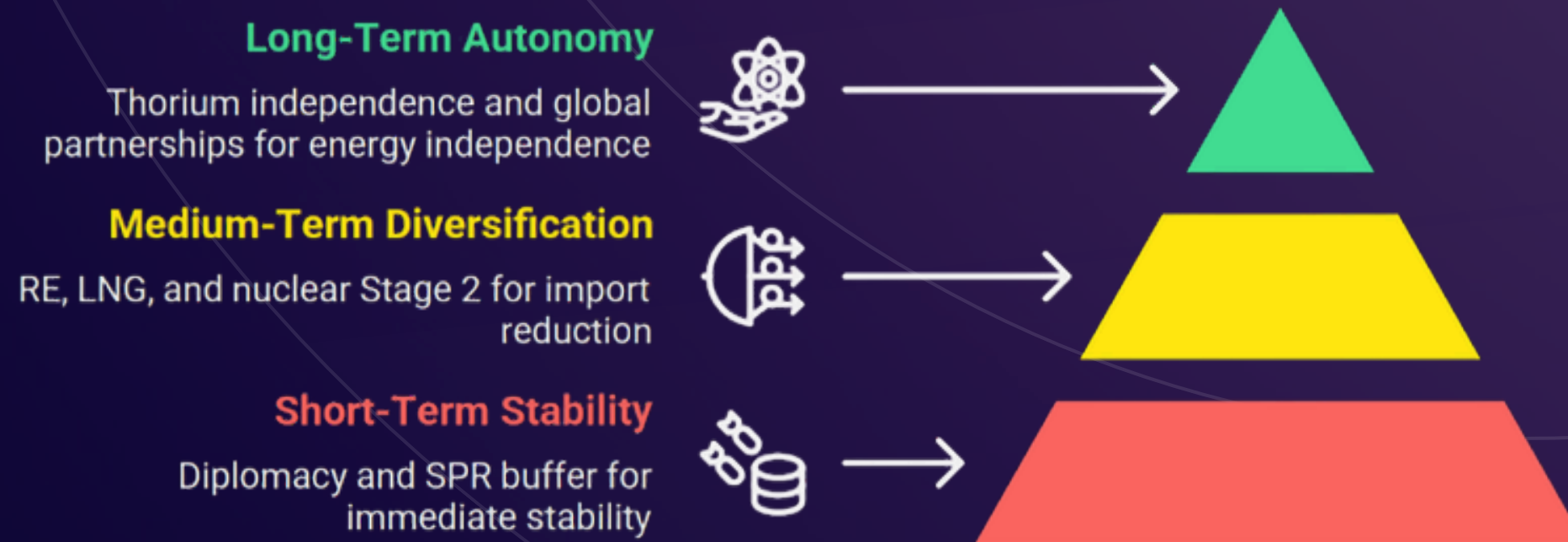
- Fully consistent with Panchamrit (1 BT CO₂ cut by 2030) & net-zero 2070.
- RE + nuclear replace imported fossil → no new coal; green hydrogen pilots for fertiliser/steel.
- Avoids lock-in; accelerates clean baseload for growth.



Thorium-Powered Independence – Partnerships, Resilience & Self-Reliance

✓ Strategic Framework:

- **Resilient Supply Chain:** Expand SPR to 90–180 day standard (from 74 days) via PPP + international joint storage (e.g., UAE/US).
- **Geopolitical Energy Diplomacy:** Deepen I2U2/QUAD/BRICS pacts; invest in African/Latin fields; protect Chabahar post-crisis.
- **Thorium-Led Nuclear Vision:** Leverage world's largest thorium reserves (~25% global) via Homi Bhabha's 3-stage program – accelerate AHWR/SMR thorium cycle for Stage 3; target 100 GW nuclear by 2047.



✓ Overall Roadmap Summary

- **Short:** Diplomacy + Russia surge + SPR drawdown/export priority → Immediate stability.
- **Medium:** RE 500 GW + diversified LNG/oil + nuclear Stage 2 → 30–40% import reduction.
- **Long:** Thorium independence + 100 GW nuclear + global partnerships → Energy autonomy by 2047, shock-proof economy.

Expected Outcomes: Inflation volatility ↓ 50%, GDP insulated, net-zero accelerated, strategic autonomy enhanced.



MUJAHEED



PRAVEEN



MUHAMMAD

Our Team

VoltEdge is a multidisciplinary team of engineering students from **MNNIT Allahabad**, focused on structured problem-solving and clear decision-making. We identify overlooked real-world challenges and translate them into practical, scalable solutions through rigorous analysis and thoughtful design.